

The entire system aims for decentralisation. It decentralises from the hierarchy, yet maintains some hierarchical support so one doesn't get flooded in a huge mass of machine protocol.

How does the vSoC provide security to the infrastructure?

One of the main highlights of a SynchronoKnot vSoC is its security engine, which caters to the network layer as well. The engine provides decentralised real-time authentication and authorisation of users to vSoC, users to virtual machines, and virtual machines to other virtual machines. All this while still maintaining LDAP, active directory, SSH (secure shell), and much more.

The decentralised blockchain identity management system uses fast blockchain cryptography in real-time to confirm user identity. Blockchain SSH allows transparent fast login to virtual machines using a standard user name. Moreover, the security engine features AuthControl, a distributed fault-tolerant system that serves as a scalable, secure and simple alternative to LDAP, active directory and other systems.

The ARPLess interstellars are offered as power modules. They enable network bifurcation and security at layer 2 across stacked or unstacked virtual LAN connections. A 28-bit interstellar identification is assigned to the virtual machine interface. Only virtual machines with the same interstellar identification can communicate with each other, irrespective of their location.

The ARPLess interstellars create a secure vacuum for trusted communication within virtual machines. They do not allow forced traffic diversion from poisoned ARP caches of virtual machines to reach



Figure 3: SynchronoKnot certified vSoC plus SoCs stacked with spacers and pillars (Image courtesy of SynchronoKnot)

undesired destinations. Plus, they ignore requests from virtual machines that impersonate the original to force diversion of traffic or gain access.

In this way, the security engine ensures that there are no chances of session hijacking.

The business opportunities in India

SynchronoKnot presents an amazing opportunity for systems integrators to provide great ROI solutions to customers. Certified systems integrators can provide SynchronoKnot solutions to their clients along with the hardware from ASRock. For being a certified systems integrator, one needs to have a clientele background in Linux. After certification, systems integrators can brand the services and sell on their own.

How does product delivery work and what is the price structure?

SynchronoKnot services have to be bought from systems integrators locally, and hardware can be ordered from the systems integrator or directly from the ASRock website.

SynchronoKnot solutions need vSoC software along with their licences. There is no requirement for physical or virtual switches, SAN/NAS distributed block storage, or a team to manage the whole infrastructure. SynchronoKnot software and hardware seamlessly control and manage all virtual machines, and no large space is required for that. This helps decrease costs significantly, while still maintaining a high level of performance, scalability, security, and overall responsiveness and control.

The base price for a SynchronoKnot vSoC licence is 100 euros, and first time registrants are required to purchase the 'Starter' package consisting of four Base TYPE-A-16 vSoC licences for 6400 euros.

One important thing to note is that SynchronoKnot servers cannot be stationed in the United States, Germany, England, Spain, Ireland, France, and Italy. However, one can have servers in any other region and provide customer service in those countries. **END** 🐧

The article is an extract from an interview with Mehul Sharma, owner of SynchronoKnot and creator of vSoC.

By: Darshil Patel

The author is a technical journalist at OSFY.